ATOM COUNT CARD

MOLE CONCEPT ACTIVITY 02

Baking Soda ($NaHCO_3$): from a printed label to moles, molecules and atoms

SECTION A - PRODUCT IDENTITY



Product: Arm & Hammer Pure Baking Soda, 755 servings per container, net weight 50g

Compound: 100% pure Sodium bicarbonate, **$NaHCO_3$**

Why is $NaHCO_3$ matters? because it acts as a chemical leavening agent that releases carbon dioxide to make baked goods rise, while also serving as a significant source of dietary sodium.

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SECTION B - BASE QUANTITATIVE DATA

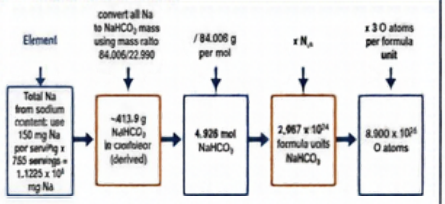
Quantity	Value	Source	Precision
Net Content	453 g	Product label	Nominal; not limiting
Concentration	100% w/w	Product label/purity	3 s.f (limits)
Density, pure baking soda(25°C)	2.20 g/cm³	Sigma-Aldrich (n.d)	3 s.f
Atomic Masses C, H, O, Na	12.011,1.008, 15.999,22.990 g/mol	CIAAW(2024)	Not limiting
Avogadro's Constant	6.022 140 76 x 10^{23} mol	BIPM(2019)	Exact
Formula	$NaHCO_3$	PubChem (n.d)	exact counts

Assumptions (labeled): 100% w/w, the conventionfor solid chemical powders;density taken at (25°C) lab usage in F is asumed.

SECTION C - MOLAR MASS BUILD-UP

Element	Atoms	gmol/each	g/mol	% by Mass
Na	1	22.990	22.990	27.367
H	1.	1..008	1..008	1.200
C	1	12.011	12.011	14.298
O	3	15.999	47.997	57.135
Total				
$NaHCO_3$	6		84.006	100.000

SECTION D - THE MOLE MAP



SECTION E - MOLE-CONVERSION CALCULATIONS

1. **Molar mass build-up:** 22.990 + 1.088 + 12.011 + 313.199 **= 84.008 g/mol**
2. **Label Bridge:** 150 mg Na/Serving x 755 servings = **113.250 mg = 1.13 x 10^5 mg** (total Na in container)
3. **Mass of Na to mass of $NaHCO_3$:** 113,250 Na x(1 g/1000 mg) x (84.006 g/22.990 g) = **413.82 g = 4.14 x 10^2 g $NaHCO_3$**
4. **Mass to moles:** 413.82 g / 84.006 g/mol = 4.9261mol = 4.93 **mol $NaHCO_3$**
5. **Moles to formula units:** 4.9261 mol x 6.022 x 10^{23}/mol = 2.9665 x 10^{24} = **2.97 x 10^{24} formula units $NaHCO_3$**
6. **Oxygen atoms in compound:** 2.9665 x 10^{24} formula units x (3 O atoms / 1 formula unit)= 8.8995 x 10^{24} = **8.8995 x 10^{23} O atoms**
7. **Percent composition (two=route check for Na)**
   Route A: 4.14^{g}g x 0.27367 = **113.30 g = 1.13 x 10^2 g Na**
   Route B: 4.93 mol x (1 mol Na/1 mol $NaHCO_3$ x 22.990 g/mol = **1.13 x 10^2 g Na**
8. **Reverse:** 10.0 mol $NaHCO_3$ x 84.006 g/mol x (1 serving/0.600 g) = 1400.1 servings = 1.40 x 10^3 servings
   > **show scale-up:** 1 container has 4.926 mol of $NaHCO_3$, 4.926 mol / 755 servings = 6.52 x 10^{-3} mol/ serving
   1.00 mol Na would need about 153 servings worth of Na equivalent (1.00 mol/6.525 10^{-3} mol = 153.26 = 153 servings)

SECTION F - SCALE-UP

1 ONE BOX	2 ONE PANTRY SHELF (12 (12 in/s)	3 ONE SCHOOL	4 REVERSE:
4.93 mol (4.926 mol = 414 g) = 0.414 kg	59.1 mol (59.11 mol) (4,966 g) = 4,97 kg	1.18 x 1.0³ mol (1, 182 mol) = 99.317 g = 99.3 kg	1.00 mol (84.006 g) = 0.0840 kg

SECTION G - SIG-FIG CHECK AND ONE-BASQUESTR AGEANT

Calculator: 2..966524 x 10^{24} formula units
Reported: 2.97 x 10^{24} formula units

- Limits: The final result must be reported to 3 significant figures based on the nutrition label, details (150 mg and 600 mg). Exact values apply to Na chemical subcripts, and serving counts.
- *453 g* has nominal packaging limits; standard mass declarations do not restrict structural chemical precision
- Guard digits kept until the last line of every calculation.

4.93 mol **$NaHCO_3$**

One bar of baking soda contains a sodium bicarbonate equal to about 4.9 moles, roughly 3 x 10^2 formula units, per pure sodium ($NaHCO_3$) (1650 mg/1.650 mol $NaHCO_3$, or about 414 grams.

SECTION H: REFERENCES (APA 7TH EDITION)

Arm & Hammer. (2026). Pure baking soda [Product label]. Church & Dwight Co., Inc.

Bureau International des Poids et Mesures. (2019). The International System of Units (SI) (9th ed.). BIPM. https://bipm.org

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PubChem compound summary for CID 516892: Sodium bicarbonate. PubChem. https://nih.gov

Sigma-Aldrich. (n.d.). Sodium bicarbonate safety data sheet. Merck KGaA. https://sigmaaldrich.com